



KUZMAN CONSULTING

BIOINFORMATICIANS AT YOUR SERVICE

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INFLAMOMX
CA24166



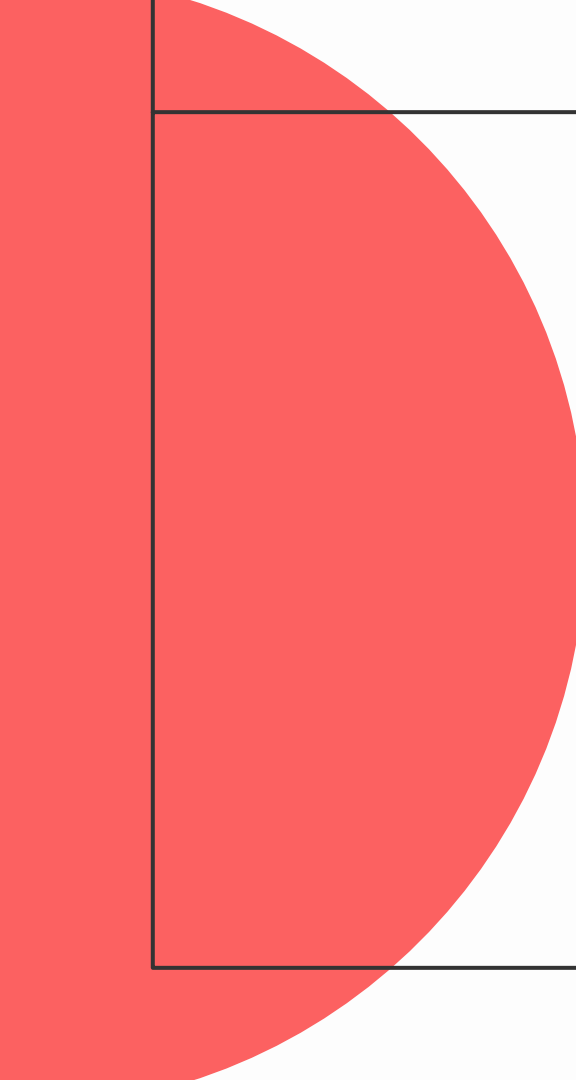
InnoThyroGen



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SVEUČILIŠTE U SPLITU
MEDICINSKI FAKULTET
UNIVERSITY OF SPLIT
SCHOOL OF MEDICINE

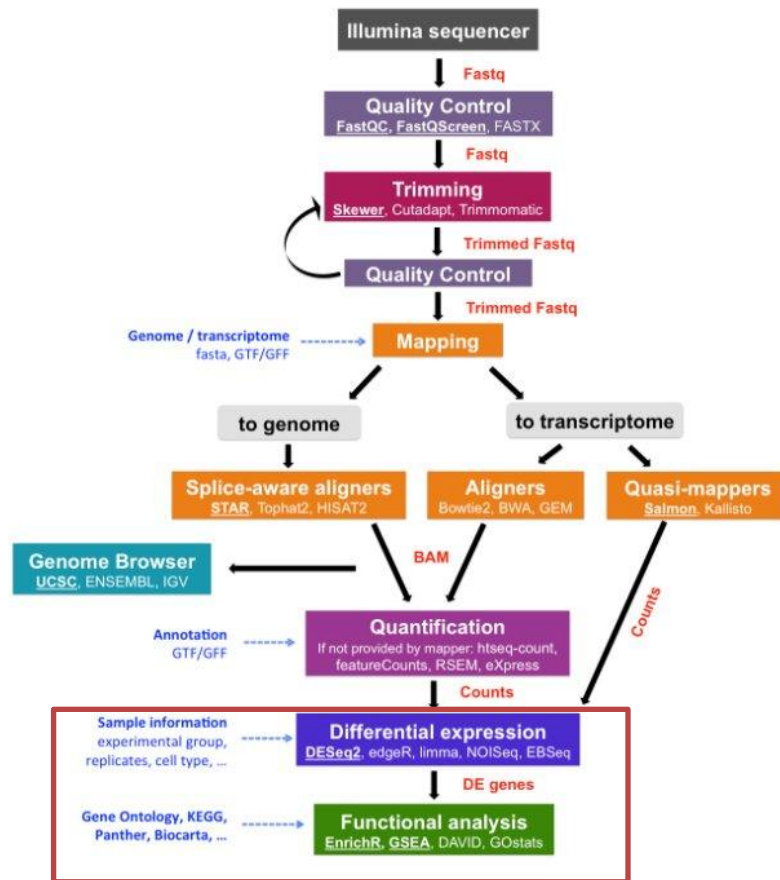
A large red circle is partially visible on the left side of the slide, overlapping the white rectangular area.

Diferencijalna ekspresija gena

RNA-seq analiza

Što nam je još ostalo?

1. Koji geni se razlikuju između različitih uvjeta (kontrola/tretman)
2. Koja je biološka interpretacija gena / signalnih puteva?



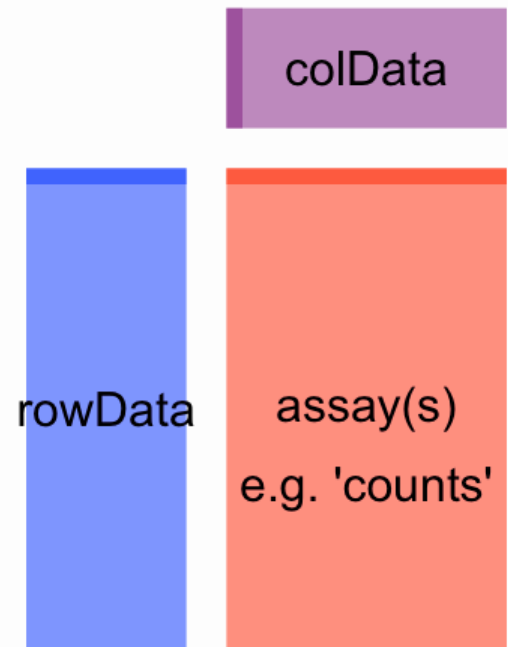
Ulazni podaci za DGE

- Broj *readova* (koliko je puta sekvenca iz nekog gena očitana) kao mjera ekspresije gena
- Što je veći broj *readova* → razina ekspresije tog gena u uzorku je viša

		Uzorci				
Geni		SRR1039508	SRR1039509	SRR1039512	SRR1039513	SRR1039516
	ENSG00000000003	679	448	873	408	1138
	ENSG00000000005	0	0	0	0	0
	ENSG00000000419	467	515	621	365	587
	ENSG00000000457	260	211	263	164	245
	ENSG00000000460	60	55	40	35	78

Ulazni podaci za DGE

1. Sirovi brojevi očitavanja (raw counts) nalaze se u assay slotu
2. Fenotipski podaci o uzorcima nalaze se u colData slotu
3. Informacije o genima nalaze se u rowData slotu



SummarizedExperiment object

Ulazni podaci za DGE

Koje je naše istraživačko pitanje?

Koje dvije (više) skupina
uspoređujemo?

design <- ~ **treatment**

design <- ~ sex + age + **treatment**

design <- ~ litter + **treatment**

design <- ~ sex + age + litter + **treatment**

	sex	age	litter	treatment
sample1	M	11	1	Ctrl
sample2	M	13	2	Ctrl
sample3	M	11	1	Treat
sample4	M	13	1	Treat
sample5	F	11	1	Ctrl
sample6	F	13	1	Ctrl
sample7	F	11	1	Treat
sample8	F	13	2	Treat

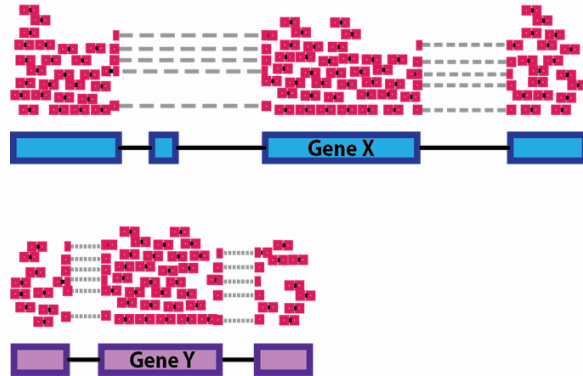
Contrasts (design)

Broj *readova* kao mjera ekspresije gena

Situacija 1:

Koji gen u istom uzorku je više eksprimiran?

Sample A Reads

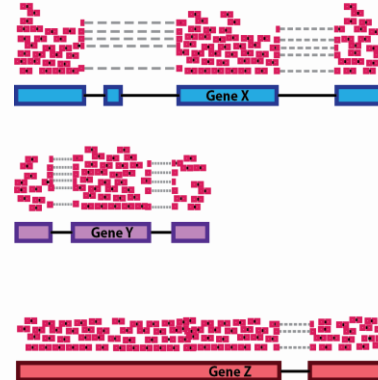


Dulji geni = više *readova*

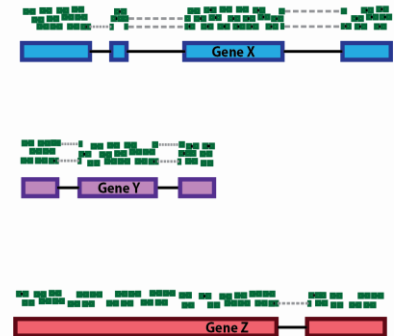
Situacija 2:

Koji uzorak ima više eksprimirane gene?

Sample A Reads



Sample B Reads



Uzorci mogu imati različit broj ukupnih
readova (veličina knjižnice)

Potreba za normalizacijom

CILJ: Koji geni su visoko eksprimirani radi biološkog razloga, a ne tehničke varijabilnosti.

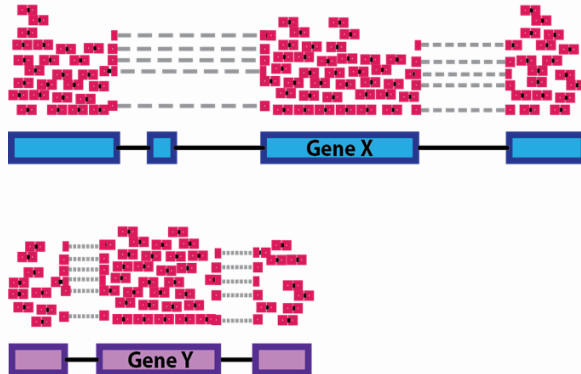
Unutar istog uzorka

Duljina gena, GC

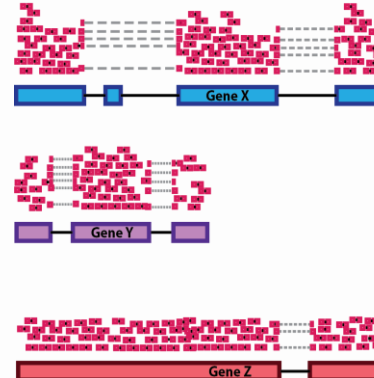
Između uzoraka

Dubina sekvenciranja

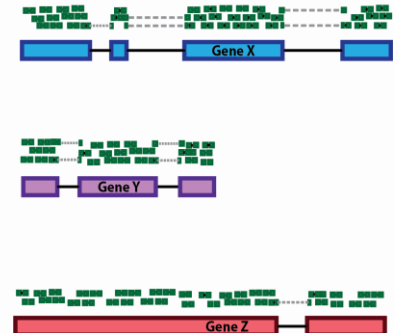
Sample A Reads



Sample A Reads



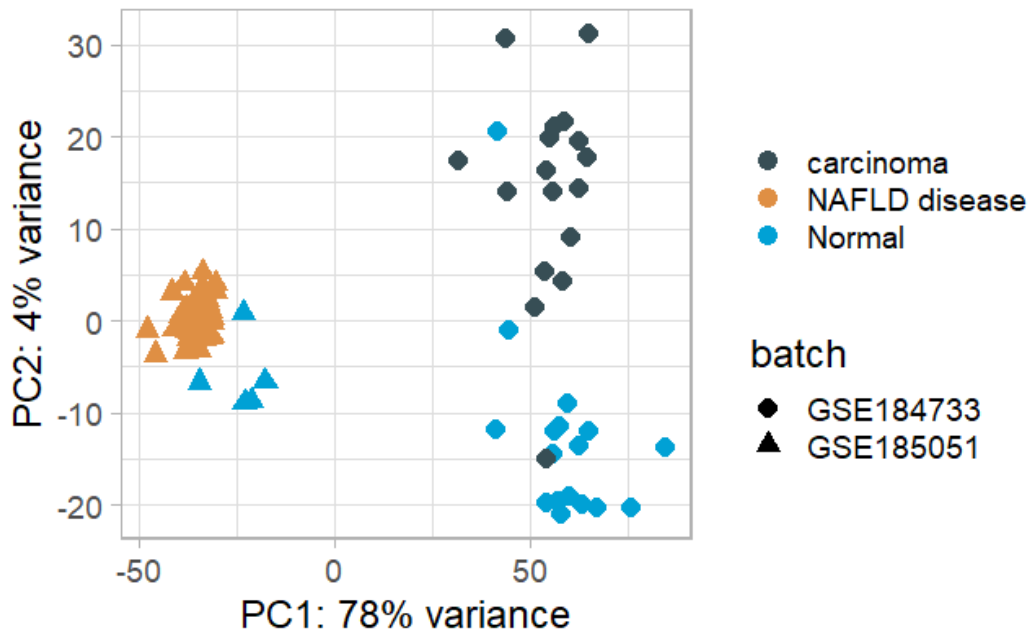
Sample B Reads



Biološka i tehnička varijabilnost

- **Biološka varijabilnost** – stvarne razlike između grupa uzoraka
- **Tehnička varijabilnost:**
 - Sekvenciranje, pipetiranje, kvaliteta RNA (nasumična varijabilnost)
 - Tzv. **batch effect** – sustavna razlika između skupina uzoraka koja proizlazi iz tehničkih razloga
- Provjera postojanja *batch effecta* – analiza glavnih komponenti (PCA)
 - Prikazuje kako se uzorci međusobno razlikuju na temelju ukupne genske ekspresije

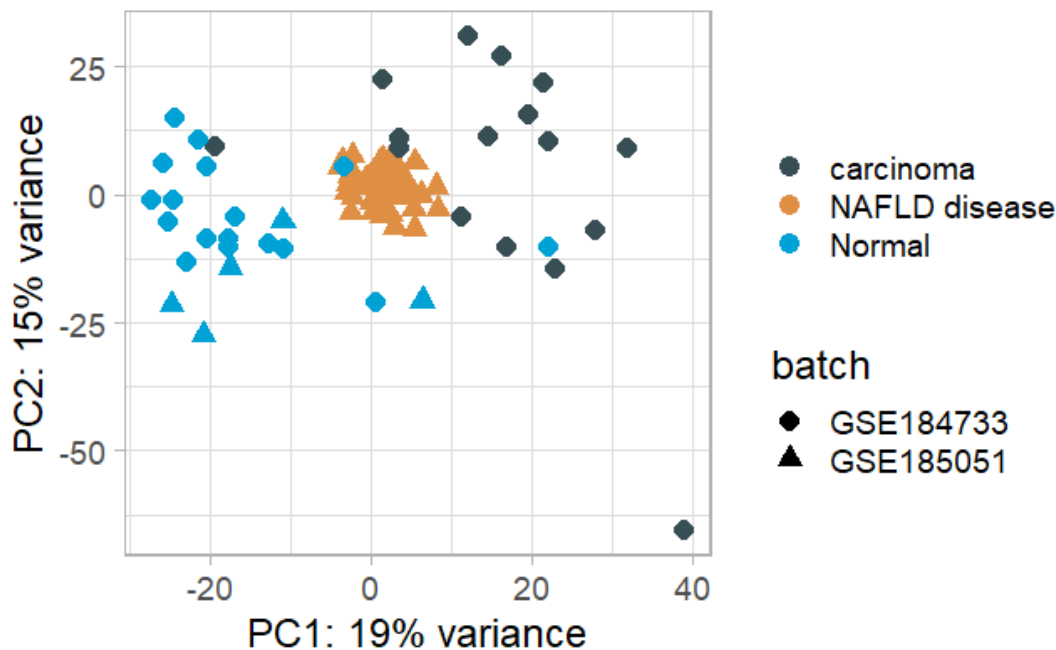
Provjera kvalitete na razini uzoraka



Što je krivo na ovom PCA plotu?

Provjera kvalitete na razini uzoraka

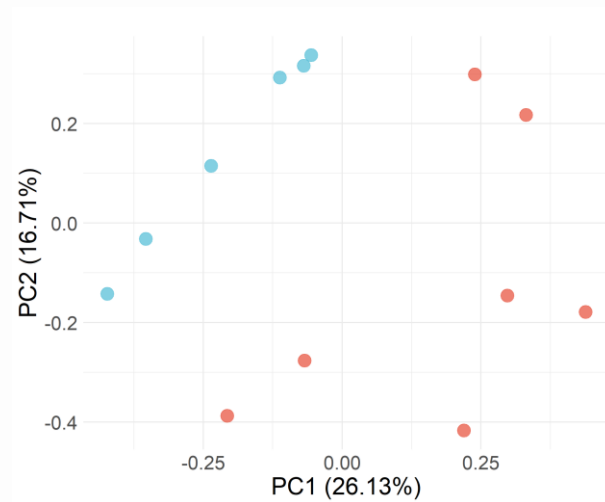
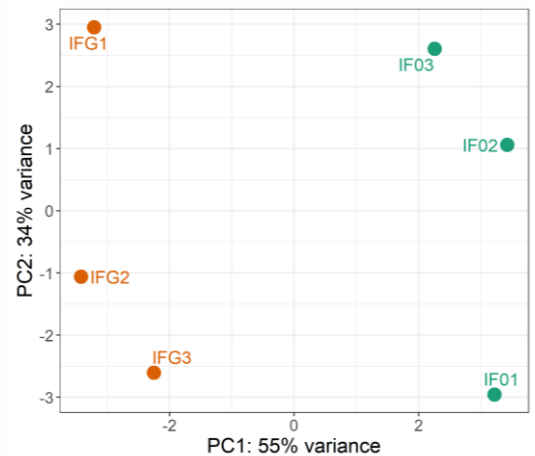
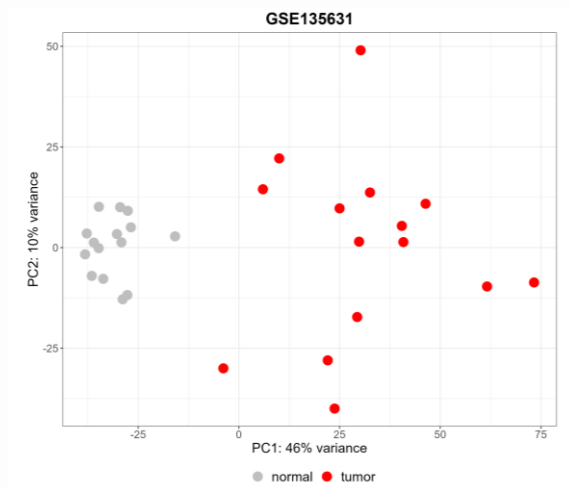
Nakon uklanjanja *batch effecta*



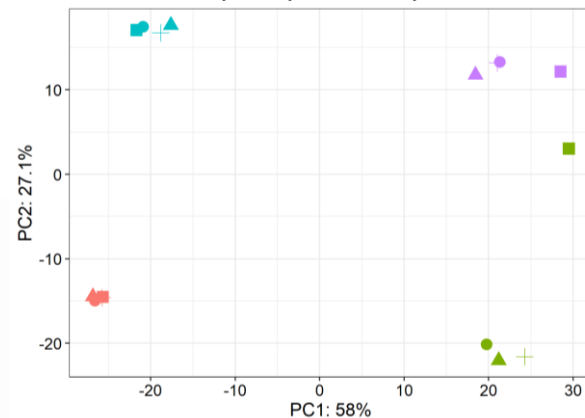
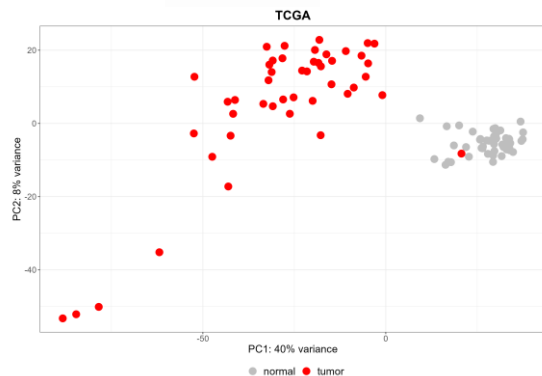
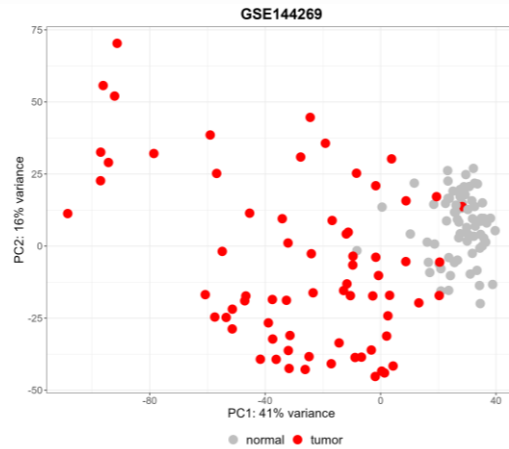
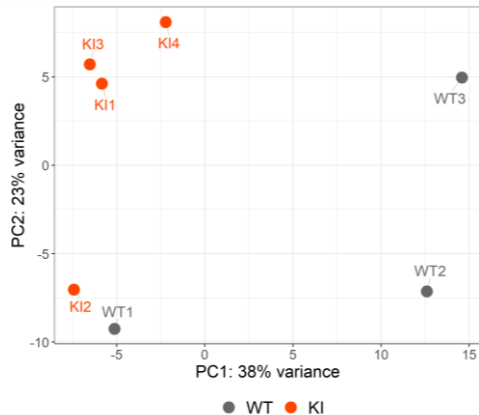
Real-life PCA 😊



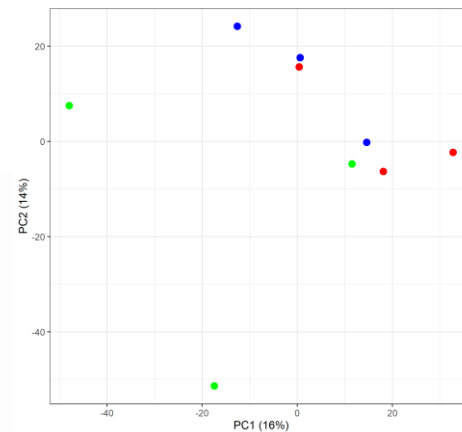
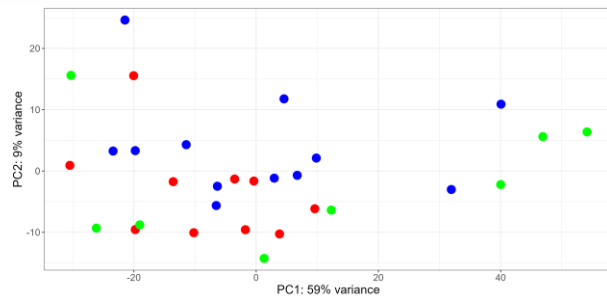
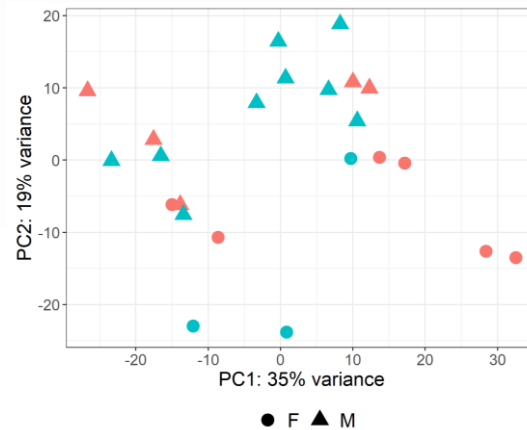
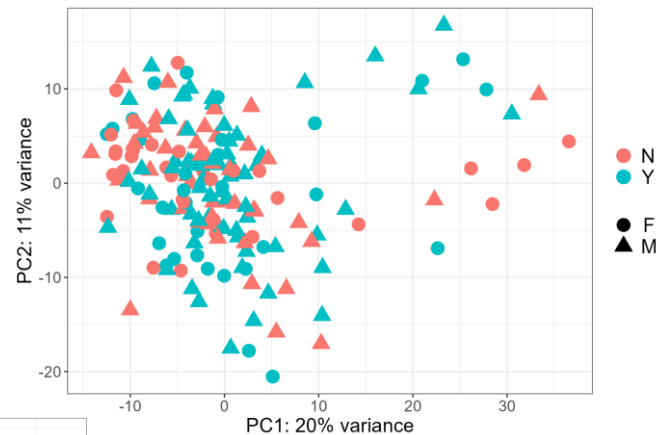
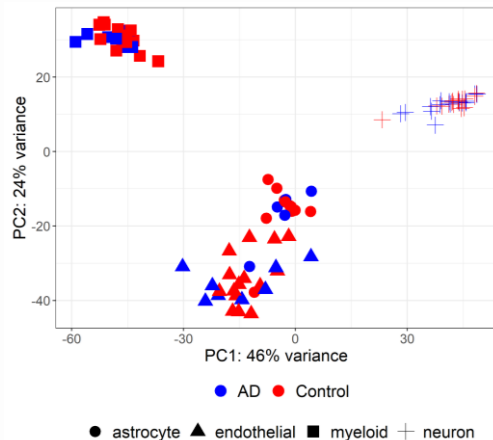
PCA: The Good



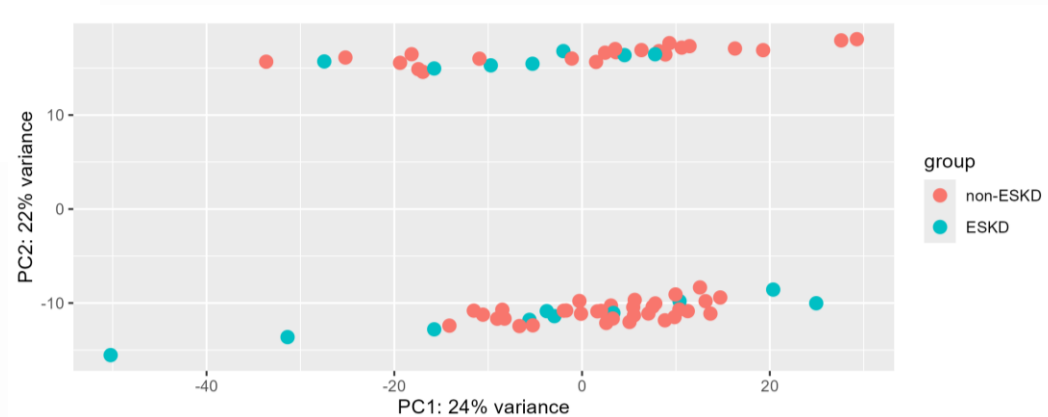
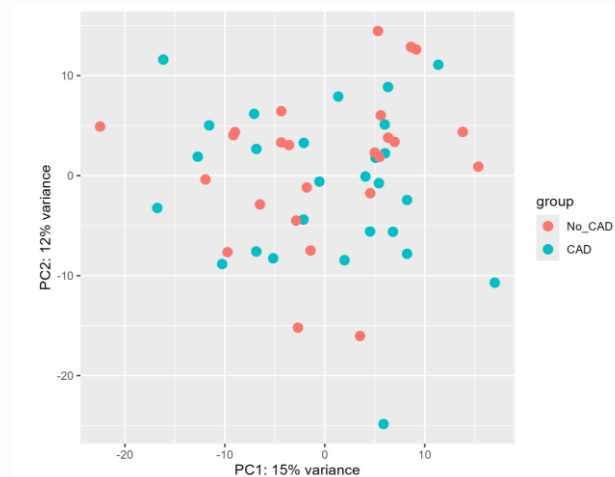
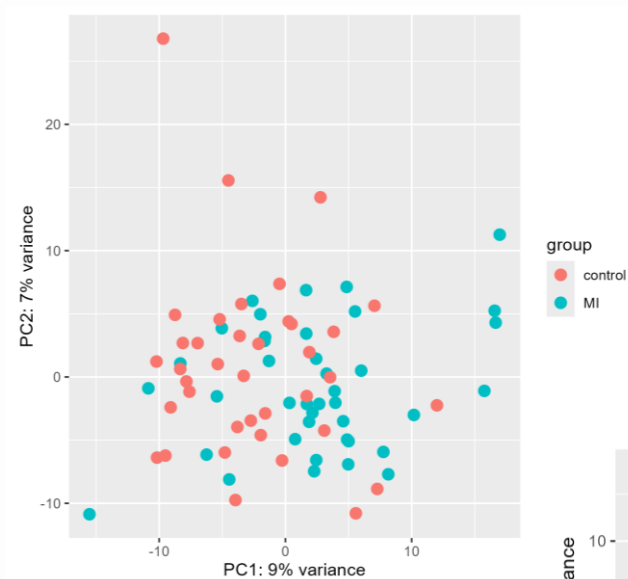
PCA: The Good(ish) / Bad



PCA: The Ugly



PCA: The Ugly



DESeq2 – najpopularniji alat za DGE

Analyzing RNA-seq data with DESeq2

Michael I. Love, Simon Anders, and Wolfgang Huber

10/29/2024

<https://bioconductor.org/packages/devel/bioc/vignettes/DESeq2/inst/doc/DESeq2.html>

Abstract

A basic task in the analysis of count data from RNA-seq is the detection of differentially expressed genes. The count data are presented as a table which reports, for each sample, the number of sequence fragments that have been assigned to each gene. Analogous data also arise for other assay types, including comparative ChIP-Seq, HiC, shRNA screening, and mass spectrometry. An important analysis question is the quantification and statistical inference of systematic changes between conditions, as compared to within-condition variability. The package DESeq2 provides methods to test for differential expression by use of negative binomial generalized linear models; the estimates of dispersion and logarithmic fold changes incorporate data-driven prior distributions. This vignette explains the use of the package and demonstrates typical workflows. [An RNA-seq workflow](#) on the Bioconductor website covers similar material to this vignette but at a slower pace, including the generation of count matrices from FASTQ files. DESeq2 package version: 1.46.0

- Standard workflow
 - Quick start
 - How to get help for DESeq2
 - Acknowledgments
 - Funding
 - Input data
 - Why un-normalized counts?
 - The DESeqDataSet
 - Transcript abundance files and *tximport* / *tximeta*
 - Tximeta for import with automatic metadata
 - Count matrix input
 - *htseq-count* input
 - *SummarizedExperiment* input
 - Pre-filtering
 - Note on factor levels
 - Collapsing technical replicates

Bioconductor
The R Project for Statistical Computing

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Home > Bioconductor 3.20 > Software Packages > DESeq2

DESeq2

This is the **released** version of DESeq2; for the devel version, see DESeq2.

Differential gene expression analysis based on the negative binomial distribution

platforms	tests	support	in BioC	builds	updated	software licenses	dependencies
3	277,226	1.25, 7, 1.34	12 years	34	75		

DOI: 10.18129/B9.bioc.DESeq2

Bioconductor version: Release (3.20)

Estimate variance-mean dependence in count data from high-throughput sequencing assays and test for differential expression based on a model using the negative binomial distribution.

Author: Michael Love [aut, cre], Constantin Ahimann-Eltze [ctb], Kwame Forbes [ctb], Simon Anders [aut, ctb], Wolfgang Huber [aut, ctb], RADIANT EU FP7 [fnd], NIH NHGRI [fnd], CZI [fnd]

Maintainer: Michael Love <michaelsalohove@gmail.com>

Citation (from within R, enter citation("DESeq2")):

Love MI, Huber W, Anders S (2014). "Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2." *Genome Biology*, 15, 550. doi:10.1186/s13059-014-0550-8

DESeq2 – filtriranje prije analize

- Uklanjanje gena koji imaju mali broj *readova* u većini uzoraka
 - Nedovoljno informacije za otkriti značajnu razliku u ekspresiji
- Izračunati zbroj po svakom retku (za svaki gen) kroz sve uzorke
- Ukloniti sve gene čiji je zbroj *readova* po retku ispod odabranog praga (< 10 očitavanja kroz sve uzorke)

	SRR1039508	SRR1039509	SRR1039512	SRR1039513	SRR1039516
ENSG000000000003	67	44	87	40	1138
ENSG000000000005	0	0	0	0	0
ENSG000000000419	467	515	621	365	587
ENSG000000000457	260	211	263	164	245
ENSG000000000460	2	5	1	0	1

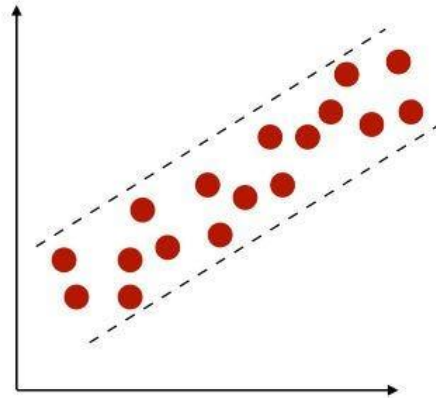
Genes with extreme count outlier

Genes with low mean normalized counts ('Independent filtering')

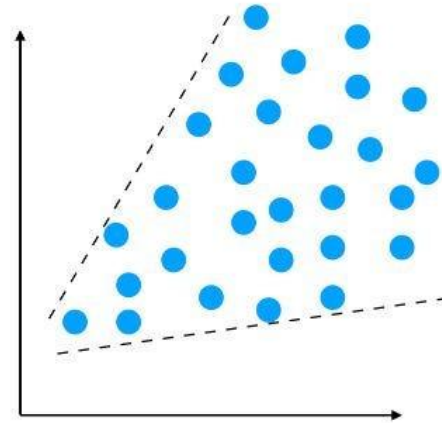
Genes with zero counts

Problem heteroskedastičnosti RNA-seq podataka

- Varijabilnost podataka nije jednaka kroz cijeli raspon vrijednosti



Homoscedasticity



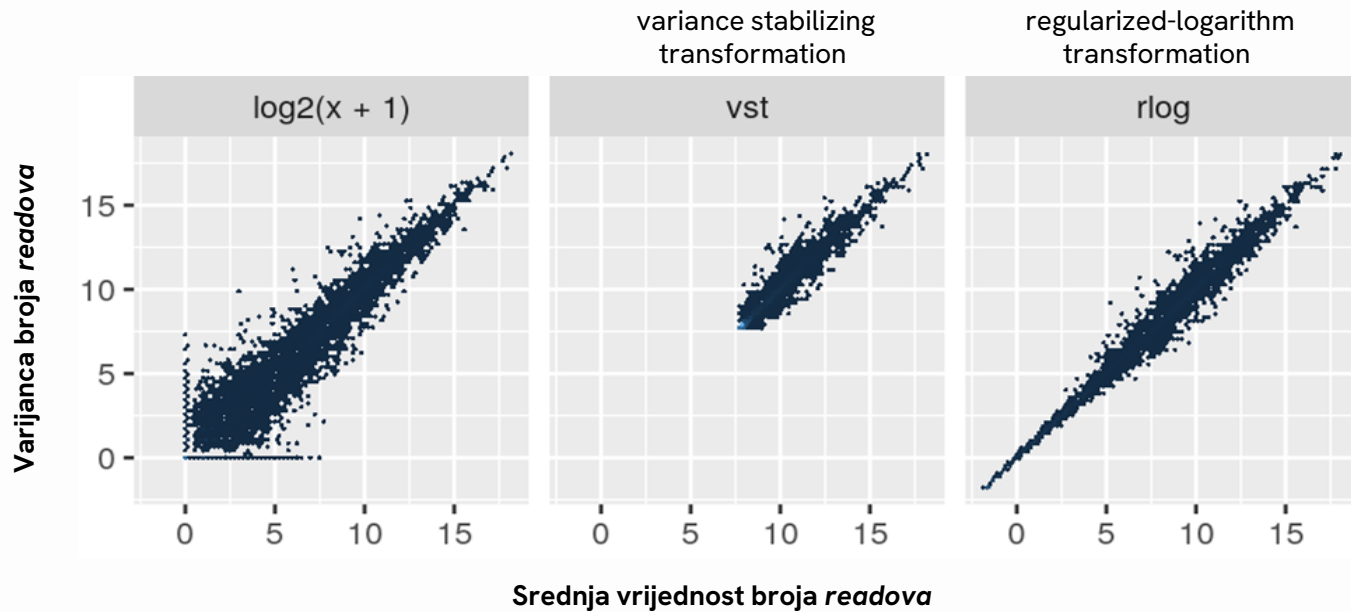
Heteroscedasticity

Problem heteroskedastičnosti RNA-seq podataka



- Slabo eksprimirani geni imaju puno više relativnog šuma
- Snažno eksprimirani geni su stabilniji
- DESeq koristi model na temelju negativne binomne distribucije
- Problem za vizualizaciju (PCA)

DESeq2 – transformacija podataka



Uklanjanje ovisnosti varijance o srednjoj vrijednosti

DESeq2 – normalizacija ovisno o dubini sekvenciranja

- DESeq2 provodi internu normalizaciju pomoću geometrijske sredine
 - Zanemaruju se jako varijabilni i/ili visoko eksprimirani geni
1. za svaki gen se izračuna geometrijska sredina broja *readova* te se to koristi kao referentna vrijednost
 2. za svaki uzorak se izračuna omjer broja *readova* svakog gena s geometrijskom sredinom
 3. medijan tih omjera po uzorku = faktor veličine (*size factor*) za svaki uzorak
 4. podijelimo broj *readova* svakog gena s faktorom veličine → normalizirani broj *readova*

gene	untreated	treated
A	2	10
B	4	12
C	6	20
D	30	0



gene	untreated	treated	Pseudo ref
A	2	10	$\text{Sqrt}(2 \cdot 10) = 4.47$
B	4	12	6.93
C	6	20	10.95
D	30	0	0



gene	untreated	treated	Pseudo ref	untreat/ref	treat/ref
A	2	10	4.47	0.45	2.24
B	4	12	6.93	0.58	1.73
C	6	20	10.95	0.55	1.83
D	30	0	0	0	0

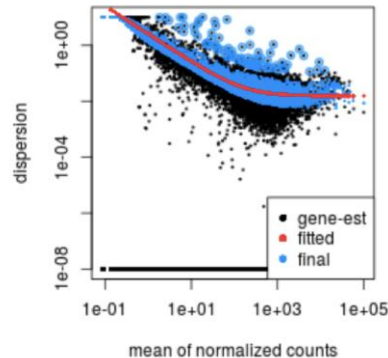
0.5 1.78



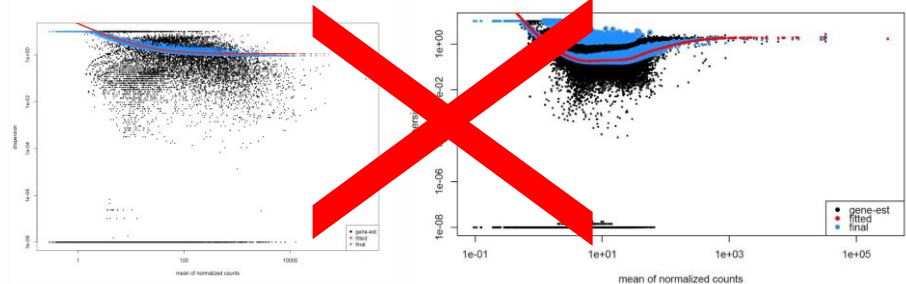
gene	norm_untreat	norm_treat
A	4.016	5.62
B	8.032	6.74
C	12.05	11.24
D	60.24	0

DESeq2 – procjena varijabilnosti između replikata

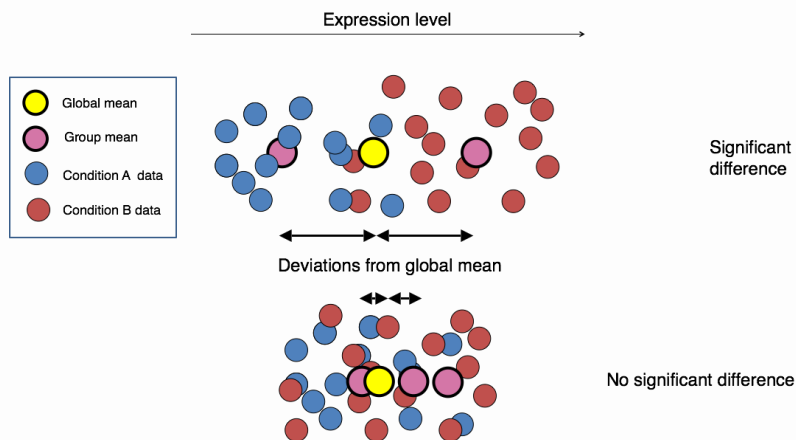
- Zbog bioloških razlika između uzoraka i tehničke varijabilnosti, broj *readova* za isti gen nikad nije identičan među replikatima → DESeq2 procjenjuje i modelira te varijabilnosti → pouzdano razlikovanje stvarnih promjena ekspresije od slučajnih odstupanja
- DESeq2 pretpostavlja da geni sa sličnom razinom ekspresije također imaju i sličnu varijabilnost



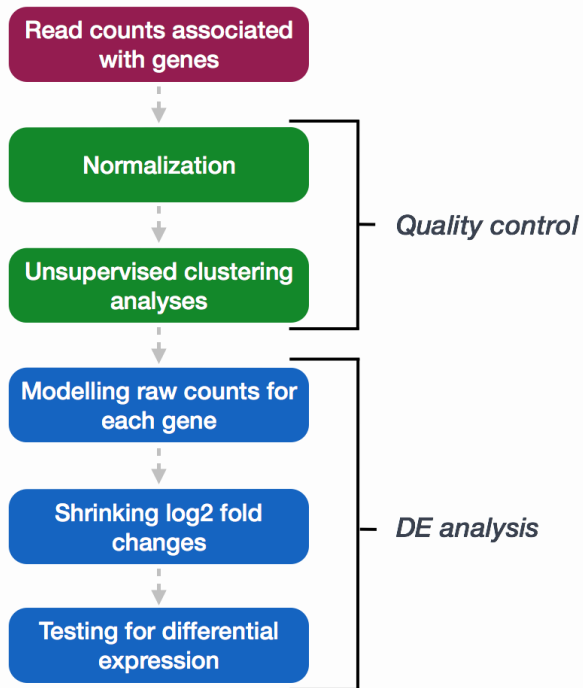
Disperzija od 0,01
znači 10% varijacije
oko očekivane
srednje vrijednosti
među biološkim
replikatima



DESeq2 – testiranje diferencijalne ekspresije



Procjena odstupanja od ukupne srednje vrijednosti broja *readova* u obje grupe



DESeq2 – testiranje diferencijalne ekspresije

Read counts
associated with genes

Normalization

Unsupervised
clustering analyses

Modeling raw counts
for each gene

Shrinking log2 fold
changes

Testing for differential
expression

Jednom jedinom funkcijom u
DESeq2-u se odrade svi ovi koraci 😊

Estimate size factors

Estimate gene-wise dispersions

Fit curve to gene-wise
dispersion estimates

Shrink gene-wise dispersion
estimates

GLM fit for each gene

Generalizirani linearni model

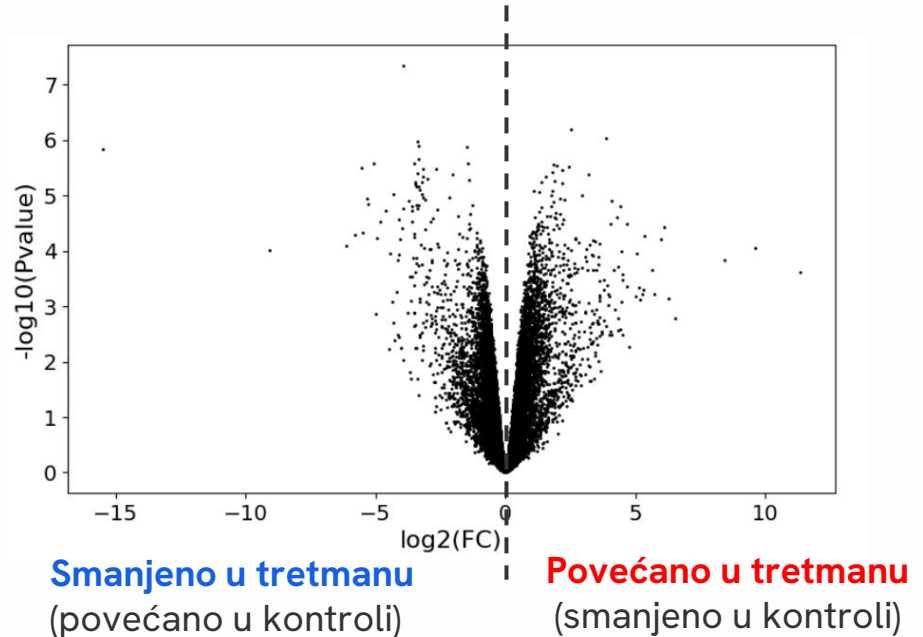
Statistički test kojim procjenjujemo
razlikuje li se razina genske ekspresije
između skupina

Fold change vs. log2 fold change

- **Fold change** - mjera koja opisuje kolika je promjena genske ekspresije između dvije skupine uzoraka
 - Tretman vs. Kontrola
 - Tretman / Kontrola

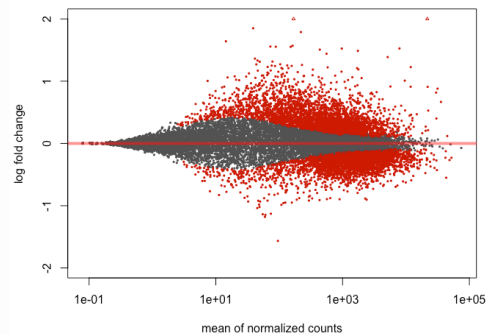
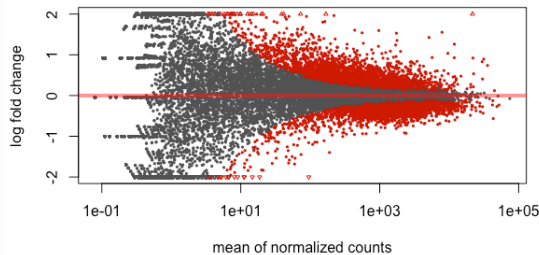
$\log_2 FC = 1$ (dvostruko veća ekspresija)

$\log_2 FC = -1$ (dvostruko manja ekspresija)



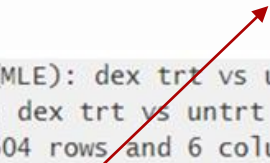
Smanjenje log2 fold change (tzv. shrinkage)

- Smanjuje varijabilnost i vuče procjenu log2 *fold changea* prema nuli kada je količina informacija za gen mala
 - Mali broj broja *readova*
 - Visoke vrijednosti disperzije
- Smanjuje broj lažno pozitivnih rezultata



DESeq2 – primjer rezultata

srednja vrijednost ekspresije u svim uzorcima



```
## log2 fold change (MLE): dex trt vs untrt
## Wald test p-value: dex trt vs untrt
## DataFrame with 31604 rows and 6 columns
##
```

	baseMean	log2FoldChange	lfcSE	stat	pvalue	padj
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
## ENSG00000000003.14	739.940717	-0.3611537	0.106869	-3.379419	0.000726392	0.00531137
## ENSG000000000419.12	511.735722	0.2063147	0.128665	1.603509	0.108822318	0.29318870
## ENSG000000000457.13	314.194855	0.0378308	0.158633	0.238479	0.811509461	0.92255697
## ENSG000000000460.16	79.793622	-0.1152590	0.314991	-0.365912	0.714430444	0.87298038
## ENSG000000000938.12	0.307267	-1.3691185	3.503764	-0.390757	0.695977205	NA
## ...	...	...	...	...	...	...
## ENSG00000285979.1	38.353886	0.3423657	0.359511	0.952310	0.340940	0.600750
## ENSG00000285987.1	1.562508	0.7064145	1.547295	0.456548	0.647996	NA
## ENSG00000285990.1	0.642315	0.3647333	3.433276	0.106235	0.915396	NA
## ENSG00000285991.1	11.276284	-0.1165515	0.748601	-0.155692	0.876275	0.952921
## ENSG00000285994.1	3.651041	-0.0960094	1.068660	-0.089841	0.928414	NA

DESeq2 – primjer rezultata

razlika u ekspresiji
 $\log_2(\text{tretman}/\text{kontrola})$

```
## log2 fold change (MLE): dex trt vs untrt
## Wald test p-value: dex trt vs untrt
## DataFrame with 31604 rows and 6 columns
##
```

	baseMean	log2FoldChange	lfcSE	stat	pvalue	padj
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
## ENSG00000000003.14	739.940717	-0.3611537	0.106869	-3.379419	0.000726392	0.00531137
## ENSG000000000419.12	511.735722	0.2063147	0.128665	1.603509	0.108822318	0.29318870
## ENSG000000000457.13	314.194855	0.0378308	0.158633	0.238479	0.811509461	0.92255697
## ENSG000000000460.16	79.793622	-0.1152590	0.314991	-0.365912	0.714430444	0.87298038
## ENSG000000000938.12	0.307267	-1.3691185	3.503764	-0.390757	0.695977205	NA
## ...	...	...	...	...	...	...
## ENSG00000285979.1	38.353886	0.3423657	0.359511	0.952310	0.340940	0.600750
## ENSG00000285987.1	1.562508	0.7064145	1.547295	0.456548	0.647996	NA
## ENSG00000285990.1	0.642315	0.3647333	3.433276	0.106235	0.915396	NA
## ENSG00000285991.1	11.276284	-0.1165515	0.748601	-0.155692	0.876275	0.952921
## ENSG00000285994.1	3.651041	-0.0960094	1.068660	-0.089841	0.928414	NA

DESeq2 – primjer rezultata

standardna greška log *fold change*a

```
## log2 fold change (MLE): dex trt vs untrt
## Wald test p-value: dex trt vs untrt
## DataFrame with 31604 rows and 6 columns
##
```

	baseMean	log2FoldChange	lfcSE	stat	pvalue	padj
##	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
## ENSG00000000003.14	739.940717	-0.3611537	0.106869	-3.379419	0.000726392	0.00531137
## ENSG000000000419.12	511.735722	0.2063147	0.128665	1.603509	0.108822318	0.29318870
## ENSG000000000457.13	314.194855	0.0378308	0.158633	0.238479	0.811509461	0.92255697
## ENSG000000000460.16	79.793622	-0.1152590	0.314991	-0.365912	0.714430444	0.87298038
## ENSG000000000938.12	0.307267	-1.3691185	3.503764	-0.390757	0.695977205	NA
## ...	...	...	...	...	...	...
## ENSG00000285979.1	38.353886	0.3423657	0.359511	0.952310	0.340940	0.600750
## ENSG00000285987.1	1.562508	0.7064145	1.547295	0.456548	0.647996	NA
## ENSG00000285990.1	0.642315	0.3647333	3.433276	0.106235	0.915396	NA
## ENSG00000285991.1	11.276284	-0.1165515	0.748601	-0.155692	0.876275	0.952921
## ENSG00000285994.1	3.651041	-0.0960094	1.068660	-0.089841	0.928414	NA

DESeq2 – primjer rezultata

Wald testna statistika & nekorigirana p-vrijednost

```
## log2 fold change (MLE): dex trt vs untrt
## Wald test p-value: dex trt vs untrt
## DataFrame with 31604 rows and 6 columns
##
```

	baseMean	log2FoldChange	lfcSE	stat	pvalue	padj
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
## ENSG00000000003.14	739.940717	-0.3611537	0.106869	-3.379419	0.000726392	0.00531137
## ENSG000000000419.12	511.735722	0.2063147	0.128665	1.603509	0.108822318	0.29318870
## ENSG000000000457.13	314.194855	0.0378308	0.158633	0.238479	0.811509461	0.92255697
## ENSG000000000460.16	79.793622	-0.1152590	0.314991	-0.365912	0.714430444	0.87298038
## ENSG000000000938.12	0.307267	-1.3691185	3.503764	-0.390757	0.695977205	NA
## ...	...	...	...	...	...	...
## ENSG000000285979.1	38.353886	0.3423657	0.359511	0.952310	0.340940	0.600750
## ENSG000000285987.1	1.562508	0.7064145	1.547295	0.456548	0.647996	NA
## ENSG000000285990.1	0.642315	0.3647333	3.433276	0.106235	0.915396	NA
## ENSG000000285991.1	11.276284	-0.1165515	0.748601	-0.155692	0.876275	0.952921
## ENSG000000285994.1	3.651041	-0.0960094	1.068660	-0.089841	0.928414	NA

DESeq2 – primjer rezultata

Korigirana p-vrijednost

```
## log2 fold change (MLE): dex trt vs untrt
## Wald test p-value: dex trt vs untrt
## DataFrame with 31604 rows and 6 columns
##
```

	baseMean	log2FoldChange	lfcSE	stat	pvalue	padj
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
## ENSG00000000003.14	739.940717	-0.3611537	0.106869	-3.379419	0.000726392	0.00531137
## ENSG000000000419.12	511.735722	0.2063147	0.128665	1.603509	0.108822318	0.29318870
## ENSG000000000457.13	314.194855	0.0378308	0.158633	0.238479	0.811509461	0.92255697
## ENSG000000000460.16	79.793622	-0.1152590	0.314991	-0.365912	0.714430444	0.87298038
## ENSG000000000938.12	0.307267	-1.3691185	3.503764	-0.390757	0.695977205	NA
## ...	...	...	...	...	...	...
## ENSG00000285979.1	38.353886	0.3423657	0.359511	0.952310	0.340940	0.600750
## ENSG00000285987.1	1.562508	0.7064145	1.547295	0.456548	0.647996	NA
## ENSG00000285990.1	0.642315	0.3647333	3.433276	0.106235	0.915396	NA
## ENSG00000285991.1	11.276284	-0.1165515	0.748601	-0.155692	0.876275	0.952921
## ENSG00000285994.1	3.651041	-0.0960094	1.068660	-0.089841	0.928414	NA

Zašto korekcija p-vrijednosti?

1 statistički test $\rightarrow p < 0,05$?

20.000 statističkih testova $\rightarrow p < 0,05$?

Problem višestrukog testiranja hipoteza:

Veliki broj gena koji se uspoređuje $\rightarrow$ veća vjerojatnost da slučajne razlike izgledaju kao stvarno značajne $\rightarrow$ korekcija p-vrijednosti

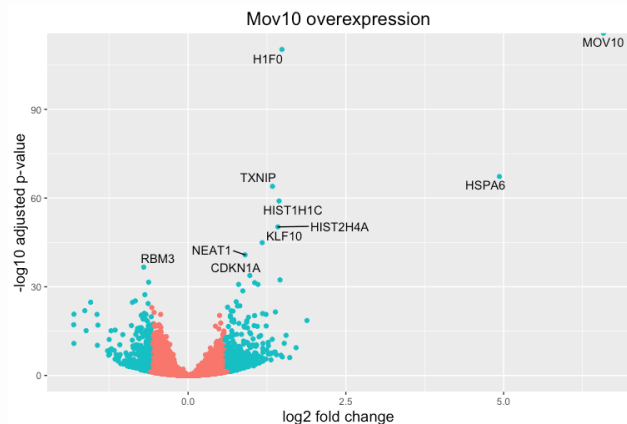
1. Bonferroni korekcija ($0,05 / \text{broj testova}$)
2. **Benjamini-Hochberg metoda / *false discovery rate* (FDR)** (procjena koji udio gena je proglašen značajnim slučajno = postotak gena za koje možemo očekivati da su lažno pozitivni)

DESeq2 – primjer rezultata (sažetak)

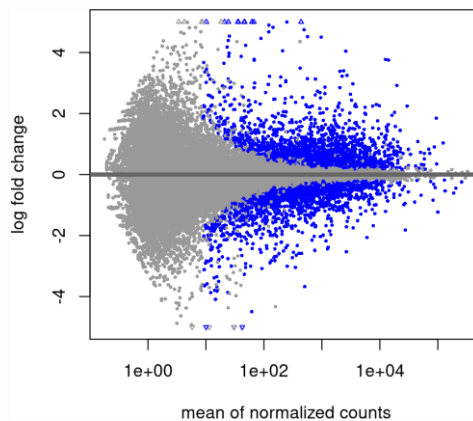
```
out of 20927 with nonzero total read count
adjusted p-value < 0.05
LFC > 0 (up)           : 6239, 30%
LFC < 0 (down)         : 5770, 28%
outliers [1]           : 0, 0%
low counts [2]          : 0, 0%
(mean count < 9)
```

DESeq2 – vizualizacija rezultata

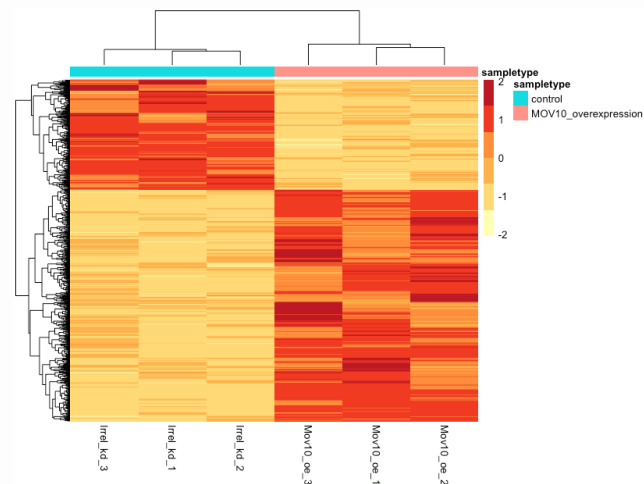
Volcano plot



MA plot

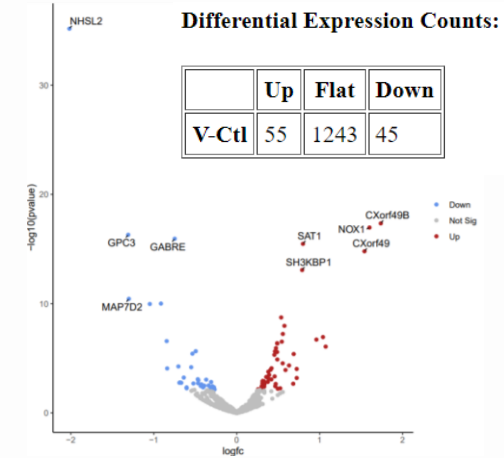


Heatmap



Što nakon DGE?

```
## log2 fold change (MLE): dex trt vs untrt
## Wald test p-value: dex trt vs untrt
## DataFrame with 31604 rows and 6 columns
##      baseMean log2FoldChange lfcSE      stat      pvalue      padj
##      <numeric>      <numeric> <numeric> <numeric> <numeric> <numeric>
## ENSG00000000003.14 739.940717 -0.3611537 0.106869 -3.379419 0.000726392 0.00531137
## ENSG000000000419.12 511.735722 0.2063147 0.128665 1.603509 0.108822318 0.29318870
## ENSG000000000457.13 314.194855 0.0378308 0.158633 0.238479 0.811509461 0.92255697
## ENSG000000000460.16 79.793622 -0.1152590 0.314991 -0.365912 0.714430444 0.87298038
## ENSG000000000938.12 0.307267 -1.3691185 3.503764 -0.390757 0.695977205 NA
## ...
## ENSG00000285979.1 38.353886 0.3423657 0.359511 0.952310 0.340940 0.600750
## ENSG00000285987.1 1.562508 0.7064145 1.547295 0.456548 0.647996 NA
## ENSG00000285990.1 0.642315 0.3647333 3.433276 0.106235 0.915396 NA
## ENSG00000285991.1 11.276284 -0.1165515 0.748601 -0.155692 0.876275 0.952921
## ENSG00000285994.1 3.651041 -0.0960094 1.068660 -0.089841 0.928414 NA
```



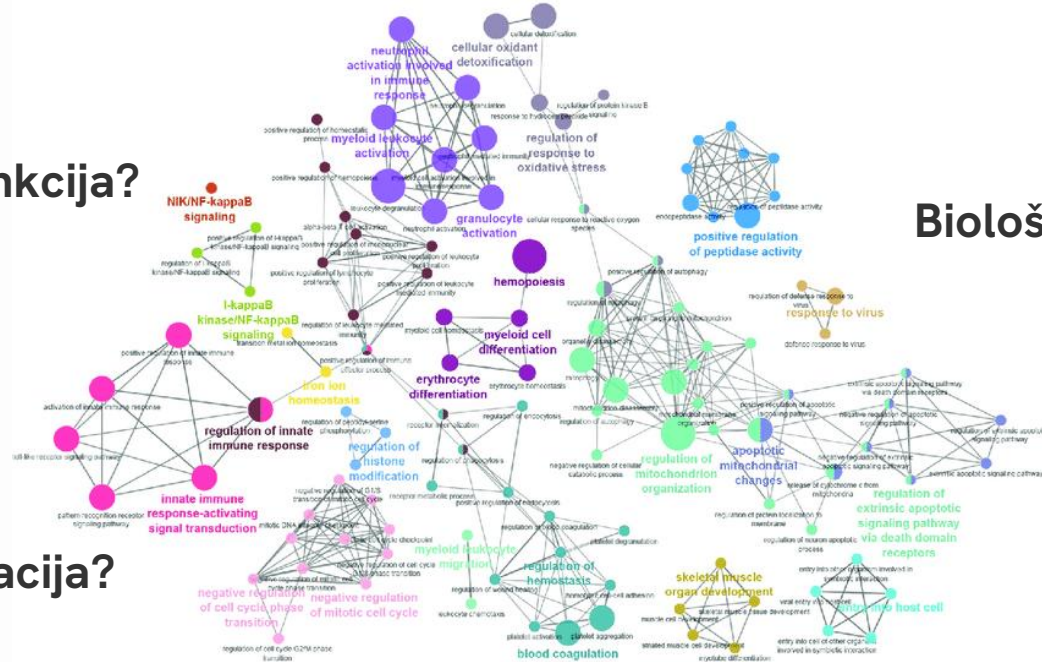
Odabir relevantnih diferencijalno eksprimiranih gena (DEGs):

- Značajna diferencijalna ekspresija ($FDR < 0,05$)
- Prag za log2 fold change (minimalno 2x veća/manja ekspresija)
- Samo geni koji kodiraju proteine

Što nakon DGE?

Funkcija?

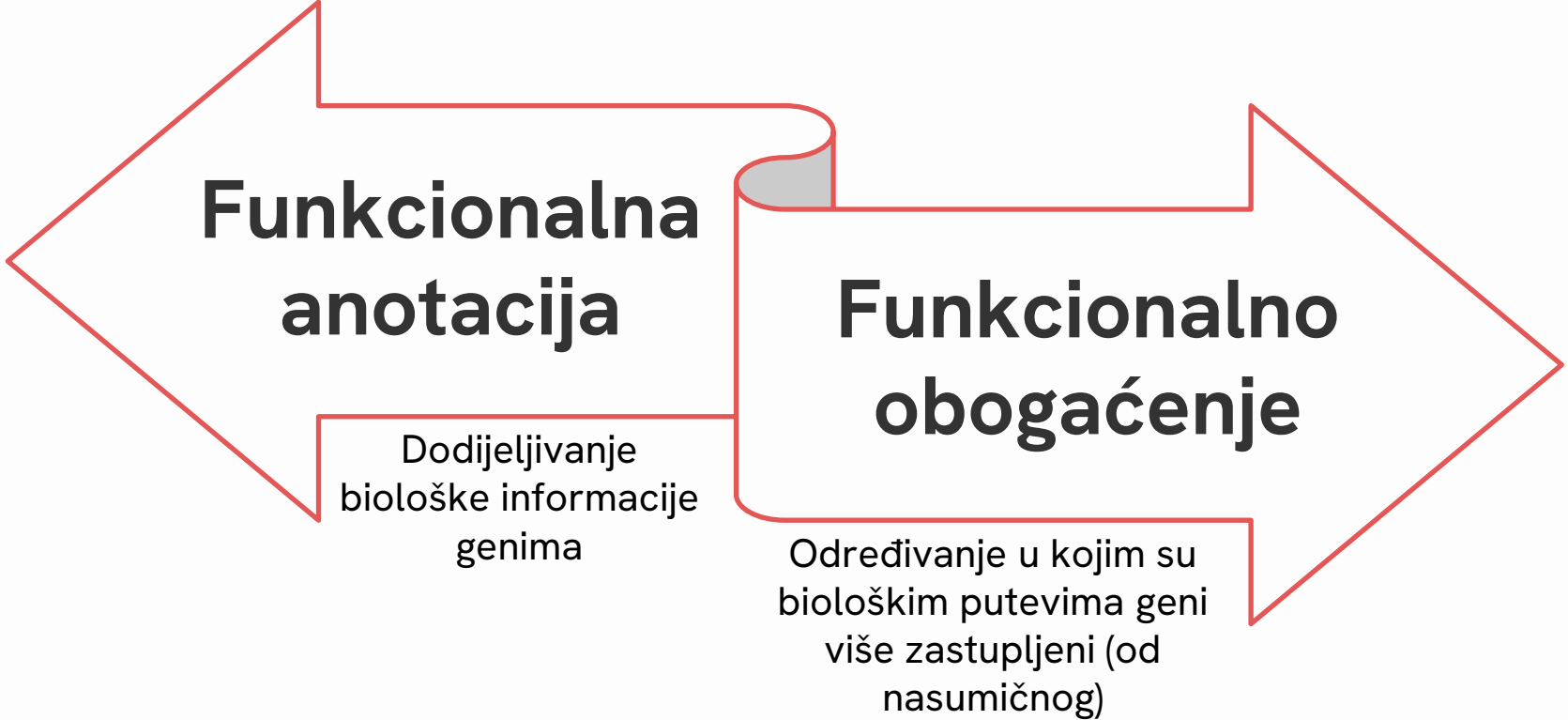
Biološki putevi?



Stanična lokalizacija?

Moramo procijeniti funkcionalno obogaćenje rezultata

Što nakon DGE?



**Funkcionalna
anotacija**

Dodijeljivanje
biološke informacije
genima

**Funkcionalno
obogaćenje**

Određivanje u kojim su
biološkim putevima geni
više zastupljeni (od
nasumičnog)

Baza podataka – Gene Ontology (GO)

Hijerarhija bioloških procesa (BP), molekularnih funkcija (MF) i staničnih komponenti (CC).

THE GENE ONTOLOGY RESOURCE

The mission of the GO Consortium is to develop a comprehensive, **computational model of biological systems**, ranging from the molecular to the organism level, across the multiplicity of species in the tree of life.

The Gene Ontology (GO) knowledgebase is the world's largest source of information on the functions of genes. This knowledge is both human-readable and machine-readable, and is a foundation for computational analysis of large-scale molecular biology and genetics experiments in biomedical research.

Search GO term or Gene Product in AmiGO ...

Any ● Ontology ● Gene Product

GO Enrichment Analysis ?

Powered by PANTHER

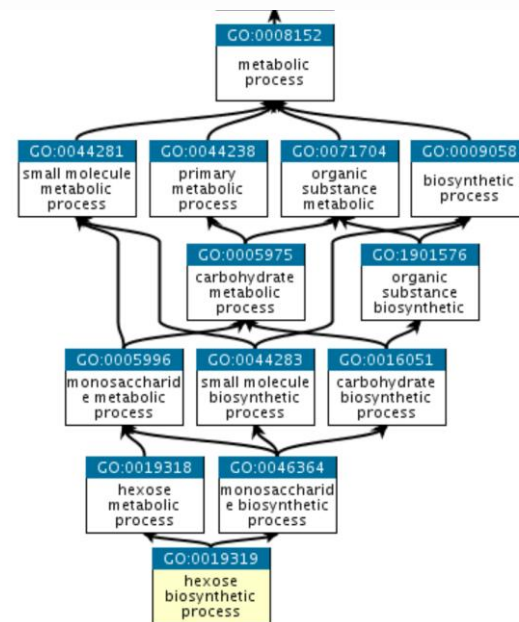
Your gene IDs here...

biological process

Homo sapiens

Examples Launch

Hint: works with UniProt ID/ACs, Gene Names, Gene Symbols, MOD IDs, Ensembl IDs, and more



Baza podataka – GO Slim

GO slim je pojednostavljena verzija Gene Ontology klasifikacije koja pruža sažet pregled funkcija gena kroz šire i općenitije kategorije.

The screenshot displays the QuickGO website interface. At the top, the header includes the QuickGO logo and the text "Gene Ontology and GO Annotations". Below the header, there are links for "Help", "Contact", "API", and "Basket". The main section features a large "Search" bar with a "Search" button. Below the search bar, an example query is provided: "e.g apoptosis; GO:0006915; ECO:0000314; tropomyosin". The interface is divided into two main panels: "View GO Annotations" and "Explore biology". The "View GO Annotations" panel shows a diagram of a "Protein/Complex/RNA" node connected to three categories: "molecular function", "biological process", and "cellular component". The "Explore biology" panel, which is circled in red, shows a flowchart for using GO terms (slims) to describe an area of interest. It includes a "Choose your terms" box, a "Protein/Complex/RNA" node, and a "Get GO slim annotations" box, all connected by arrows.

QuickGO Gene Ontology and GO Annotations

Help Contact API Basket

Search

e.g apoptosis; GO:0006915; ECO:0000314; tropomyosin

View GO Annotations

Protein/Complex/RNA

molecular function

biological process

cellular component

Explore biology

Use sets of GO terms (slims) that describe your area of interest

Protein/Complex/RNA

Choose your terms

Get GO slim annotations

Baza podataka – Reactome pathway

Detaljno kurirani ljudski signalni putovi, bogat funkcionalni kontekst.

The screenshot displays the Reactome website interface. At the top, the Reactome logo is on the left, and navigation links for Citation, Analysis, Tour, and Layout are on the right. A dropdown menu shows 'Pathways for: Homo sapiens'. Below the navigation bar, the 'Event Hierarchy' sidebar on the left lists various biological processes, each with a blue icon and a plus sign. The main content area features a search bar and a large, complex network diagram of pathways. The diagram is a hierarchical tree structure with many nodes and connecting lines, representing the relationships between different biological processes. The bottom of the page has a navigation bar with tabs for Description, Molecules, Structures, Expression, Analysis, and Downloads.

reactome 3.7 96

Pathways for: Homo sapiens

Citation: Analysis: Tour: Layout:

Event Hierarchy:

- Autophagy
- Cell Cycle
- Cell-Cell communication
- Cellular responses to stimuli
- Chromatin organization
- Circadian clock
- Developmental Biology
- Digestion and absorption
- Disease
- DNA Repair
- DNA Replication
- Drug ADME
- Extracellular matrix organization
- Gene expression (Transcription)
- Hemostasis
- Immune System

Search for a term, e.g. pten ...

Immune System

Drug ADME

DNA Replication

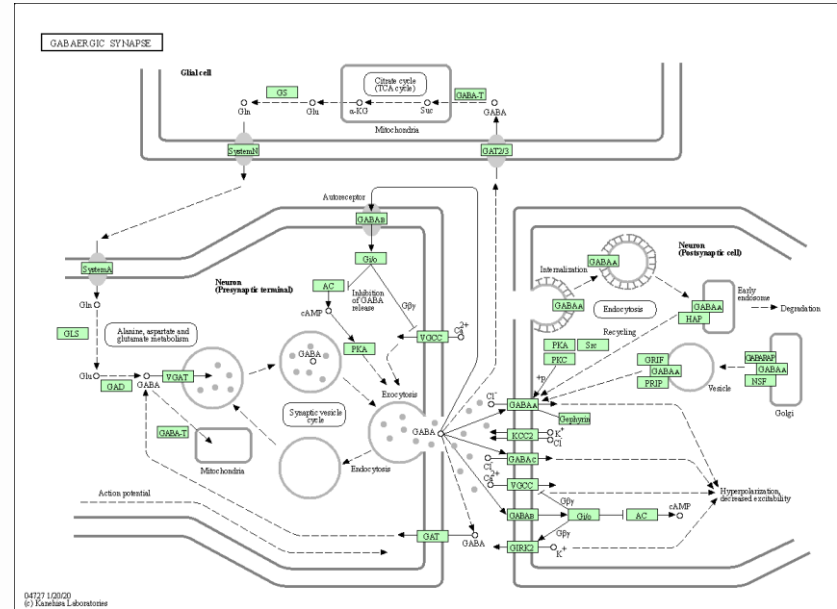
- Cell Cycle
- Programmed Cell Death
- Digestion and absorption
- Circadian clock
- DNA Repair
- Transport of small molecules
- Reproduction
- Cellular responses to stimuli
- Developmental Biology
- Signal Transduction
- Metabolism
- Sensory Perception
- Organelle biogenesis and maintenance
- Autophagy
- Protein localization
- Extracellular matrix organization
- Neuronal System
- Hemostasis
- Gene expression (Transcription)
- Disease
- Metabolism of proteins
- Vesicle-mediated transport
- Cell-Cell communication

Description Molecules Structures Expression Analysis Downloads

Baza podataka – KEGG pathway

Zbirka ručno izrađenih mapa bioloških puteva koje prikazuju naše znanje o molekularnim interakcijama, reakcijama i mrežama odnosa.

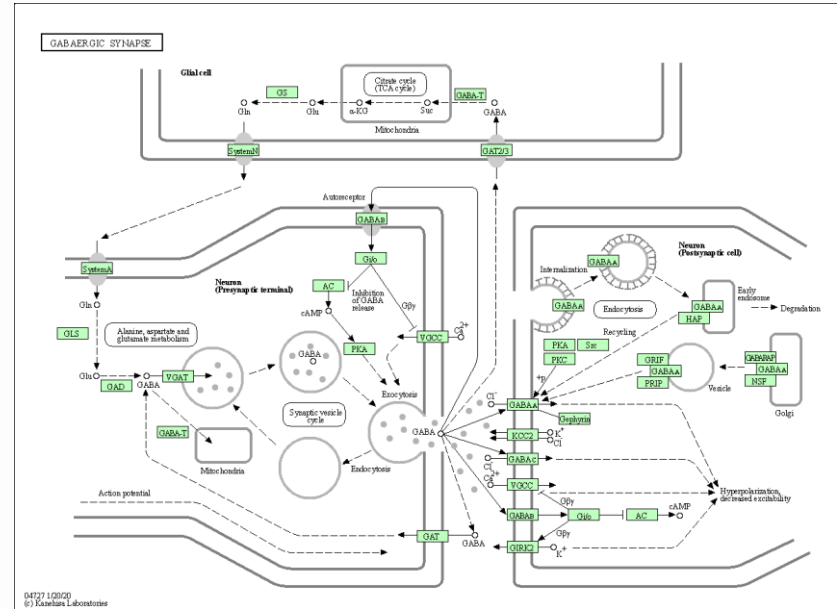
1. Metabolism
2. Genetic Information Processing
3. Environmental Information Processing
4. Cellular Processes
5. Organismal Systems
6. Human Diseases
7. Drug Development



Baza podataka – KEGG pathway

Zbirka ručno izrađenih mapa bioloških puteva koje prikazuju naše znanje o molekularnim interakcijama, reakcijama i mrežama odnosa.

1. Metabolism
2. Genetic Information Processing
3. Environmental Information Processing
4. Cellular Processes
5. Organismal Systems
6. Human Diseases
7. Drug Development



Baza podataka – The Molecular Signatures Database (MSigDB)

Velika zbirka setova gena koji obuhvaćaju regulaciju, signalizaciju, transkripcijske faktore, imunologiju i još mnogo mnogo toga.

The screenshot displays the MSigDB website. At the top, the GSEA logo and navigation links (GSEA Home, Downloads, Molecular Signatures Database, Documentation, Contact, Team) are visible. The left sidebar contains a 'MSigDB Home' section with links to Human Collections (About, Browse, Search, Investigate, Gene Families) and Mouse Collections (About, Browse, Search, Investigate), along with a Help link. Below this, the UC San Diego and Broad Institute logos are shown. The main content area is titled 'Molecular Signatures Database' and includes an 'Overview' section with a description of the database and a list of features: Examine a gene set and its annotations, Browse gene sets by name or collection, Search for gene sets by keyword, Investigate gene sets (with sub-points for Compute overlaps, Categorize members, View the expression profile, and Investigate the gene set in the online biological network repository NDEX), and Download gene sets. A 'License Terms' section at the bottom states that GSEA and MSigDB are available for use under the provided license terms. The 'Human Collections' section is a grid of 8 boxes, each describing a specific type of gene set: H (hallmark gene sets), C5 (ontology gene sets), C1 (positional gene sets), C6 (oncogenic signature gene sets), C2 (curated gene sets), C7 (immunologic signature gene sets), C3 (regulatory target gene sets), and C8 (cell type signature gene sets). A final box, C4, describes computational gene sets.

GSEA
Gene Set Enrichment Analysis

GSEA Home Downloads **Molecular Signatures Database** Documentation Contact Team

MSigDB Home

- Human Collections
 - About
 - Browse
 - Search
 - Investigate
 - Gene Families
- Mouse Collections
 - About
 - Browse
 - Search
 - Investigate
- Help

UC San Diego

BROAD INSTITUTE

Molecular Signatures Database

Overview

The Molecular Signatures Database (MSigDB) is a resource of tens of thousands of annotated gene sets for use with GSEA software, divided into Human and Mouse collections. From this web site, you can

- ▶ **Examine** a gene set and its annotations. See, for example, the [HALLMARK_APOPTOSIS human gene set page](#).
- ▶ **Browse** gene sets by name or collection.
- ▶ **Search** for gene sets by keyword.
- ▶ **Investigate** gene sets:
 - ▶ **Compute overlaps** between your gene set and gene sets in MSigDB.
 - ▶ **Categorize** members of a gene set by gene families.
 - ▶ **View the expression profile** of a gene set in a provided public expression compendia.
 - ▶ Investigate the gene set in the online [biological network repository NDEX](#)
- ▶ **Download** gene sets.

License Terms

GSEA and MSigDB are available for use under [these license terms](#).

Human Collections

H hallmark gene sets are coherently expressed signatures derived by aggregating many MSigDB gene sets to represent well-defined biological states or processes.	C5 ontology gene sets consist of genes annotated by the same ontology term.
C1 positional gene sets corresponding to human chromosome cytogenetic bands.	C6 oncogenic signature gene sets defined directly from microarray gene expression data from cancer gene perturbations.
C2 curated gene sets from online pathway databases, publications in PubMed, and knowledge of domain experts.	C7 immunologic signature gene sets represent cell states and perturbations within the immune system.
C3 regulatory target gene sets based on gene target predictions for microRNA seed sequences and predicted transcription factor binding sites.	C8 cell type signature gene sets curated from cluster markers identified in single-cell sequencing studies of human tissue.
C4 computational gene sets defined by mining large collections of cancer-oriented expression data.	

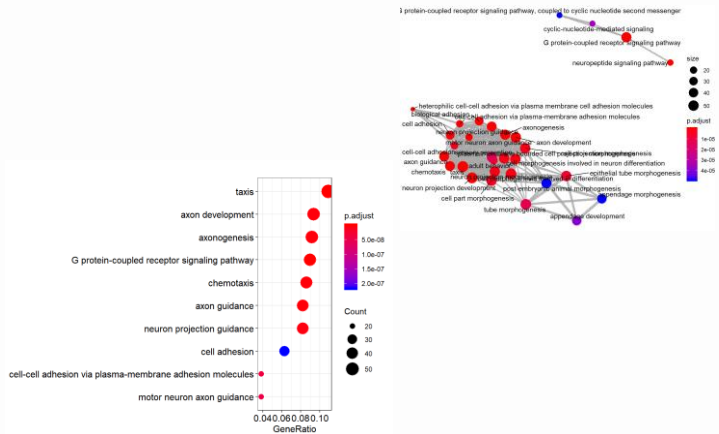
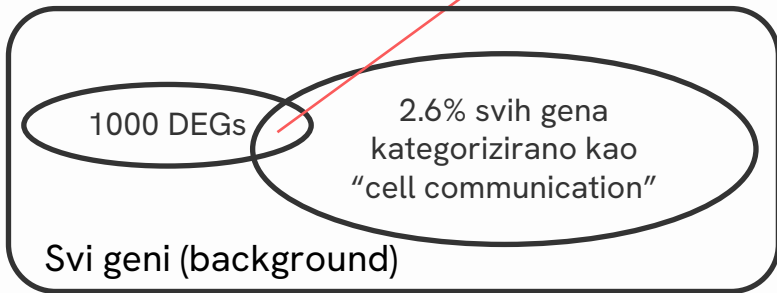
Analiza prekomjerne zastupljenosti (over-representation analysis, ORA)

Statistička metoda koja određuje jesu li geni iz unaprijed definiranih skupova (npr. oni koji pripadaju određenom GO terminu ili KEGG putu) prisutni češće nego što bi se očekivalo (prekomjerno zastupljeni) **u podskupu podataka (up/down DEGs).**

Statistički značajan preklap?

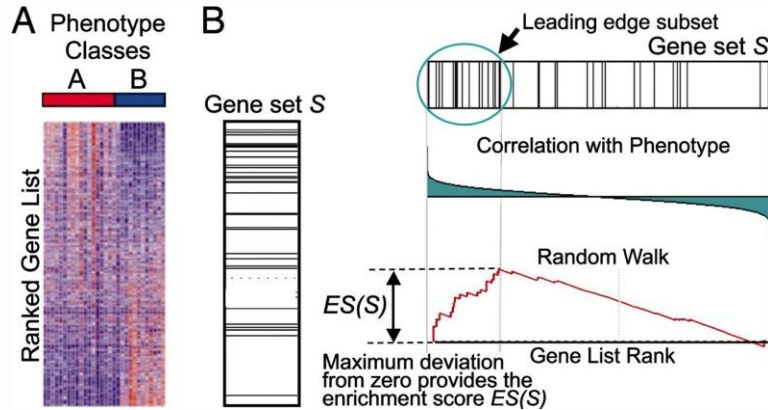
Hipergeometrijski test

20% of these DEGs are categorized as "cell communication"

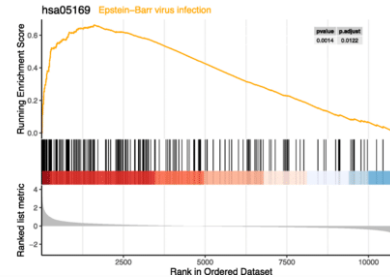


Analiza obogaćenja genskih setova (gene set enrichment analysis, GSEA)

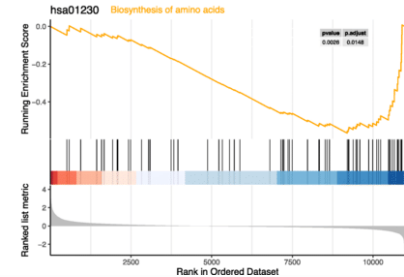
Statistička metoda koja procjenjuje jesu li unaprijed definirani genski skupovi (npr. oni koji pripadaju određenom GO terminu ili KEGG putu) statistički obogaćeni na vrhu ili dnu rangirane liste gena prema njihovoj ekspresiji (**uzima u obzir sve gene**; rangira ih ovisno o *fold changeu*).



**Aktivirani
putevi**



**Suprimirani
putevi**



Kako provesti analize obogaćenja?

Nebrojeno mnogo online alata

Protein-protein interactions
(STRING)

<https://string-db.org/>

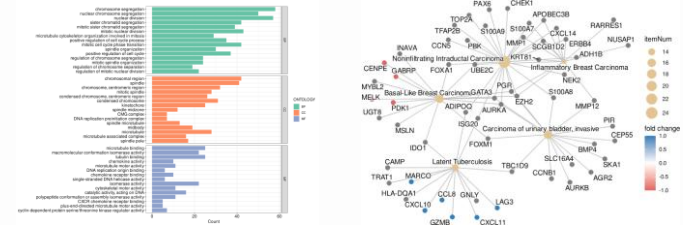
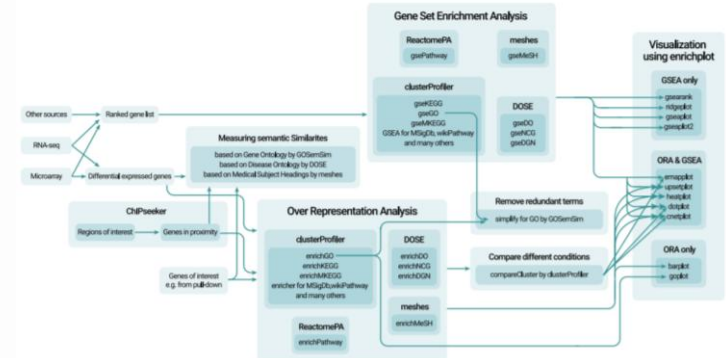
Prednost: omogućuje vizualizaciju i analizu mreža proteinskih interakcija integriranjem poznatih i predviđenih interakcija iz više izvora.

Protein ANALYSIS THrough Evolutionary Relationships
(PANTHER)

<http://www.pantherdb.org/>

Prednost: omogućuje funkcionalnu klasifikaciju (anotaciju) gena i analizu obogaćenja; ima integriran GO Slim bazu podataka.

R packet clusterProfiler



DESeq2 u praksi!



Galaxy Training!

End-to-End Analysis [link](#)

These tutorials take you from raw sequencing reads to pathway analysis

Lesson	Slides	Hands-on
1: RNA-Seq reads to counts		
collections mouse QC		
2: RNA-seq counts to genes		
limma-voom mouse QC		
3: RNA-seq genes to pathways		
mouse		

Visualisation

Tutorials covering data visualisation

Lesson	Slides	Hands-on
Visualization of RNA-Seq results with Volcano Plot		
Visualization of RNA-Seq results with heatmap2		

